Fostering STEM Engagement in Spatial Reasoning from Parent-Child Interaction: A Remote Study

Sarah L. Kiefer, Gabriella Modesti, Sam McHugh, David M. Sobel
Department of Cognitive and Psychological Sciences, Brown University

Parent-directedness during an at-home spatial reasoning activity reduces 3-to-5-year-old children's engagement in related challenges.

Background

- Parents support children's **STEM learning** through everyday activities before they begin formal education[1-3]
- Parents also foster children's interest through the *autonomy* they provide during STEM activities, which promotes children's **STEM engagement**[4]
- We investigate how parent-child interaction, focused on autonomy support during an **at-home** activity focused on spatial relations, relates to preschoolers' engagement in spatial reasoning challenges
- We predict that when parents provide children with ***less autonomy*** during the activity**, children's engagement would be reduced** during related STEM challenges.

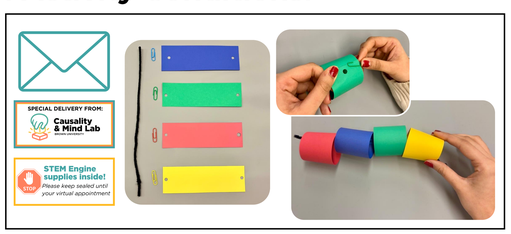
Methods

Participants: 3- to 5-year-olds (N=160, 85 girls, M_{age}= 53.87 months, SD =10.01)

PREREGISTERED

Parent-Child STEM Activity Conditions:

- **Trains** (n = 100): "Make a train" kit mailed to families. Constructed a train using craft materials. Followed by reflections.



- **Transfer** (n = 30): Families completed an alternative STEM activity that demonstrates how soap breaks surface tension in water.
- **Baseline** (n = 30) No parent-child activity

Coding PCI Style[4-5] for *Trains* Condition

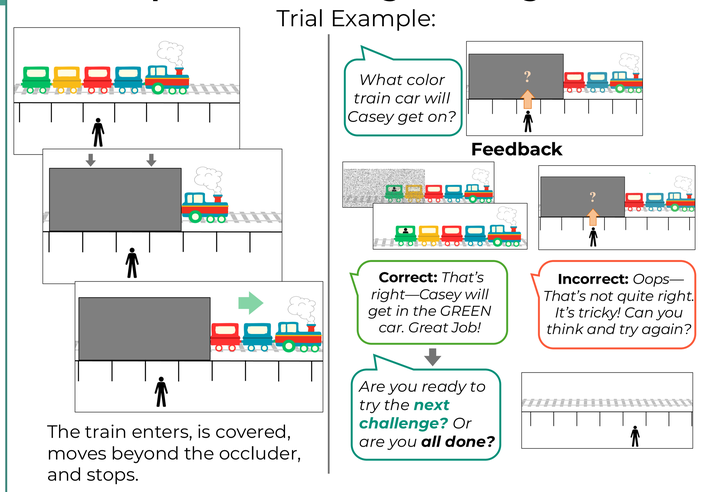
Parent-Directed	Jointly Directed	Child-Directed
Parents set goals and often take over tasks for their child	Parents supported, collaborative goal-setting	Parents let children set goals for themselves

Additional measures:

Reflections: **(1)** Causal Language and **(2)** MLU	**IC** Inhibitory control (Day/Night Stroop)[6]	**WM** Working Memory - (Digit Span)[7]	**ToM** Theory of Mind Scales[8]	**Surgency and Control** Temperament subscales (Parent report)[9]	Parent Attitudes Towards Science[10]

Spatial Reasoning Challenges

Trial Example:



The train enters, is covered, moves beyond the occluder, and stops.

Trials progressively increased in difficulty.

We measured:

- **Engagement Score:** number of challenges children participated in voluntarily (out of a possible 10)
- **Performance Score:** proportion of challenges solved correctly.

Results

- On average, children chose to complete 5.62 challenges (SD = 3.58, Range: 1-10).
- Children's engagement and performance scores were negatively correlated, $r_s(158) = -.57$, $p < .001$.

Correlations among engagement and performance scores, age, and control measures:

					Parent Questionnaire		
	Age	WM	IC	ToM	Surgency	Control	Parent Science Attitudes
Engagement	**.19***	**.25****	.11	**.20***	-.14	**.20***	.02
Performance	**.22****	**.19***	**.22****	**.18***	.05	-.09	.02

*<.05, **<.01

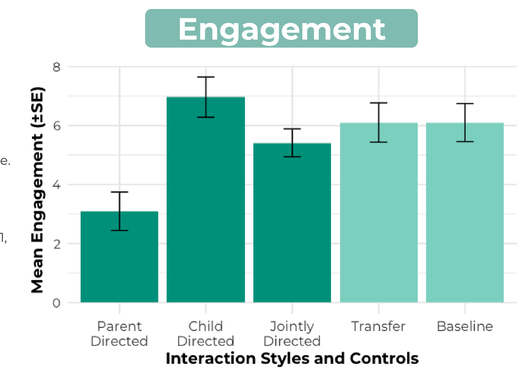
Results: Model Analyses

Children in the parent-directed group completed *fewer* challenges than those in all other interaction style groups and control conditions.

GLM, Engagement ~ Age (months) + Condition/PCI + Performance. Plot Displays SE Error bars.

- Condition effect is present in light of a main effect of **age**, B = 0.06, Wald χ^2 (1) = 11.38, $p < .001$, and **performance**, B = -6.67, Wald χ^2 (1) = 49.06, $p < .001$.

Condition effect remains significant in an exploratory model including WM, TOM, and Temperament.

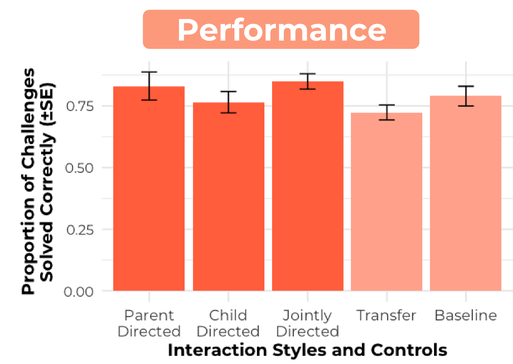


There were no significant condition differences in children's performance scores.

GLM, Performance ~ Age (months) + Condition/PCI + Engagement. All comparisons between parent-directed and other groups, n.s.

- Children were more accurate with **age**, B = 0.07, Wald χ^2 (1) = 16.02, p < .001.
- More accurate children also tended to have *lower* **engagement**, B = -0.45, Wald χ^2 (1) = 65.25, p < .001. Plot Displays SE Error bars.

An exploratory model including WM, IC, and ToM as covariates was not significant.



PCI Style Comparisons

- Children in the **child-directed group were significantly older** than those in the parent-directed and jointly-directed groups.
- We reproduced our analyses only on the PCI groups, including causal language, MLU, and our cognitive measures.
  - Only the dyad's **PCI style** significantly predicted their **engagement** score, Wald $\chi^2(2) = 7.75$, p = .02, Parent-directed < Child and Jointly-directed.
  - Overall model on **performance** was n.s., but suggested a main effect of **WM**.

Discussion

- When parents are **overly directive** in setting goals for their preschool-age child during at-home STEM activities, children are **less engaged** in related activities, following this interaction.
- This work contributes to a greater understanding of motivational influences on children's engagement with spatial reasoning, an important precursor to engineering, before children are introduced to engineering concepts in the classroom.

References

[1] Braham et al. (2018). *Dev. Psych.* **[2]** Tamis-LeMonda et al. (2004). *Child Dev.* **[3]** Morris et al. (2023). *J. Applied Dev. Psych* **[4]** Sobel (2023). *Cur Dir Psych Sci* **[5]** Fung & Callanan (2013). Poster at SRCD **[6]** Gerstadt et al. (1994). *Cognition* **[7]** Wechsler (2002). WPPSI-III. **[8]** Wellman & Liu (2004). *Child Development* **[9]** Putnam & Rothbart (2009). *J. Personality Assessment.* **[10]** Richard (2019). Dissertation.

Acknowledgements and Contact

We thank all participating families and lab members for their contributions. This research was supported by NSF (2300459). Any opinions expressed here are those of authors and not NSF.
Contact: sarah_kiefer@brown.edu; Bluesky: @sarahkiefer @candmlab

Poster copy: QR Code

Fostering STEM Engagement in Spatial Reasoning from Parent-Child Interaction: A Remote Study

Sarah L. Kiefer, Gabriella Modesti, Sam McHugh, David M. Sobel
Department of Cognitive and Psychological Sciences, Brown University

Parent-directedness during an at-home spatial reasoning activity reduces 3-to-5-year-old children's engagement in related challenges.

Background

- Parents support children's **STEM learning** through everyday activities before they begin formal education[1-3]
- Parents also foster children's interest through the *autonomy* they provide during STEM activities, which promotes children's **STEM engagement**[4]
- We investigate how parent-child interaction, focused on autonomy support during an **at-home** activity focused on spatial relations, relates to preschoolers' engagement in spatial reasoning challenges
- We predict that when parents provide children with ***less autonomy*** during the activity**, children's engagement would be reduced** during related STEM challenges.

Methods

Participants: 3- to 5-year-olds (N=160, 85 girls, M_{age}= 53.87 months, SD =10.01)

PREREGISTERED

Parent-Child STEM Activity Conditions:

- **Trains** (n = 100): "Make a train" kit mailed to families. Constructed a train using craft materials. Followed by reflections.
- **Transfer** (n = 30): Families completed an alternative STEM activity that demonstrates how soap breaks surface tension in water. Followed by reflections.
- **Baseline** (n = 30)

No parent-child activity

Coding PCI Style[4-5]

Parent-Directed
Parents set goals and often take over tasks for their child

Jointly Directed
Parents supported, collaborative goal-setting

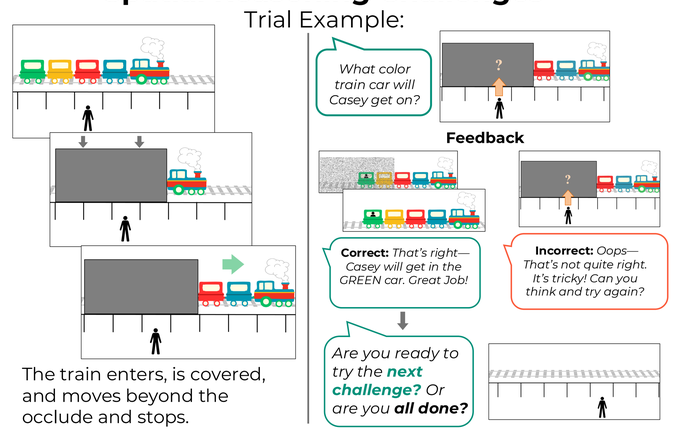
Child-Directed
Parents let children set goals for themselves

Additional measures:

Reflections: **(1)** Causal Language and **(2)** MLU	**IC** Inhibitory control (Day/Night Stroop)[6]	**WM** Working Memory - (Digit Span)[7]	**ToM** Theory of Mind Scales[8]	**Surgency and Control** Temperament subscales (Parent report)[9]	Parent Attitudes Towards Science[10]

Spatial Reasoning Challenges

Trial Example:



Trials progressively increased in difficulty.

We measured:

- **Engagement Score:** number of challenges children participated in voluntarily (out of a possible 10)
- **Performance Score:** proportion of challenges solved correctly.

Results

- On average, children chose to complete 5.62 challenges (SD = 3.58, Range: 1-10).
- Overall, on their first try, children responded correctly to 78% of their completed challenges (SD = 22%).
- Children's engagement and performance scores were correlated, $r_s(158) = -.57$, $p < .001$.

Correlations among engagement and performance scores, age, and control measures. *<.05, **<.01.

					Parent Questionnaire		
	Age	WM	IC	ToM	Surgency	Control	Parent Science Attitudes
Engagement	.19*	.25**	.11	.20*	-.14	.20*	.02
Performance	.22**	.19*	.22**	.18*	.05	-.09	.02

Results: Model Analyses

Children in the parent-directed group completed *fewer* challenges than those in all other interaction style groups and control conditions.

GLM, Engagement ~ Age (months) + Condition/PCI + Performance. Plot Displays SE Error bars.

- Condition effect is present in light of a main effect of **age**, B = 0.06, Wald χ^2 (1) = 11.38, $p < .001$, and **performance**, B = -6.67, Wald χ^2 (1) = 49.06, $p < .001$.

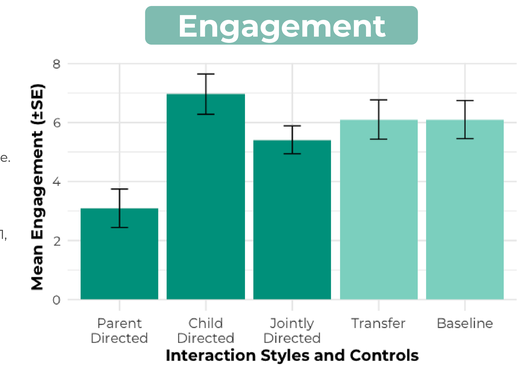
Condition effect remains significant in an exploratory model including WM, TOM, and Temperament.

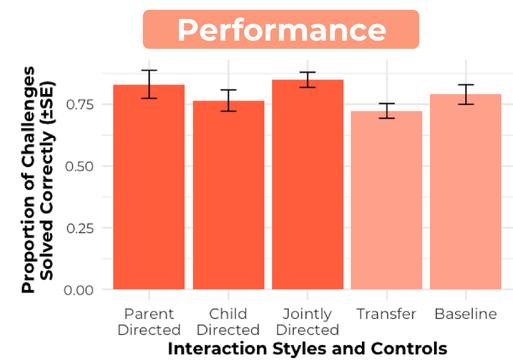
There were no significant condition differences in children's performance scores.

GLM, Performance ~ Age (months) + Condition/PCI + Engagement. All comparisons between parent-directed and other groups, n.s.

- Children were more accurate with **age**, B = 0.07, Wald χ^2 (1) = 16.02, p < .001.
- More accurate children also tended to have *lower* **engagement**, B = -0.45, Wald χ^2 (1) = 65.25, p < .001. Plot Displays SE Error bars.

An exploratory model including WM, IC, and ToM as covariates was not significant.





PCI Style Comparisons

- Children in the **child-directed group were significantly older** than those in the parent-directed and jointly-directed groups.
- We reproduced our analyses only on the PCI groups, including causal language, MLU, and our cognitive measures.
  - Only the dyad's **PCI style** significantly predicted their **engagement** score, Wald $\chi^2(2)$ = 7.75, p = .02, Parent-directed < Child and Jointly-directed.
  - Overall model on **performance** was n.s., but suggested a main effect of **WM**.

Discussion

- When parents are **overly directive** in setting goals for their preschool-age child during at-home STEM activities, children are **less engaged** in related activities, following this interaction.
- This work contributes to a greater understanding of motivational influences on children's engagement with spatial reasoning, an important precursor to engineering, before children are introduced to engineering concepts in the classroom.

References

[1] Braham et al. (2018). *Dev. Psych.* **[2]** Tamis-LeMonda et al. (2004). *Child Dev.* **[3]** Morris et al. (2023). *J. Applied Dev. Psych* **[4]** Sobel (2023). *Cur Dir Psych Sci* **[5]** Fung & Callanan (2013). Poster at SRCD **[6]** Gerstadt et al. (1994). *Cognition* **[7]** Wechsler (2002). WPPSI-III. **[8]** Wellman & Liu (2004). *Child Development* **[9]** Putnam & Rothbart (2009). *J. Personality Assessment.* **[10]** Richard (2019). Dissertation.

Acknowledgements and Contact

We thank all participating families and lab members for their contributions. This research was supported by NSF (2300459). Any opinions expressed here are those of authors and not NSF.
Contact: sarah_kiefer@brown.edu; Bluesky: @sarahkiefer @candmlab

Poster copy: QR Code